

MORTIMER: Multi-asset Optimisation and Risk Targeting with Integrated Monthly E6 Rebalancing

Nelson Alvarez, Amir Hajomar, Sean Majewski, and Miles Rack

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Summary

MORTIMER combines monthly equal-risk-contribution (ERC) weights with a causal volatility overlay across six exchange-traded funds (ETFs). In the held-out out-of-sample (OOS) test, the portfolio returned 9.52% annualised after five basis points of trading costs, with 7.47% volatility, a 0.714 Sharpe ratio, a -7.00% maximum drawdown, and annual half-L1 turnover of 1.119. Relative to matched unscaled ERC, volatility fell by 8.2% and drawdown by 9.9%, while the Sharpe ratio also declined. Validation showed larger risk reductions alongside a 15-basis-point annual return difference. The results indicate that volatility targeting moderated portfolio risk without improving risk-adjusted return.

Economic hypothesis

Investors accept equity, credit, duration, and commodity risk premia to hold exposures that diversify imperfectly across economic states. Benchmark mandates and periodic risk-budget adjustment can also leave portfolios carrying exposure after market risk has risen. Since large absolute and squared returns tend to cluster, recent returns contain information about near-term variance (Cont, 2001; Engle, 1982; Bollerslev, 1986). Realized-volatility methods formalise this persistence for forecasting (Andersen et al., 2003), and volatility-managed portfolios use risk forecasts to vary exposure through time (Moreira and Muir, 2017). MORTIMER applies this principle to a monthly, cross-asset risk-balanced portfolio: it scales risky holdings down when forecast volatility exceeds a fixed target and holds the balance in cash. The testable prediction is lower realised volatility and drawdown after trading costs, with an annual return reduction small enough to preserve risk-adjusted performance. Delayed response to abrupt shocks and foregone gains during recoveries are the corresponding economic costs.

Data and universe

The fixed universe comprises QQQ, IWM, HYG, TLT, GLD, and DBC. Yahoo adjusted closes include distributions and splits; raw closes and volume support liquidity estimates (Yahoo Finance, n.d.). Cash accrues from initial-release observations in the Archival Federal Reserve Economic Data (ALFRED) DGS3MO series (Federal Reserve Bank of St. Louis, n.d.-a, n.d.-b), using the next exchange session for availability and actual/365 day-count accrual. Factor returns use the Kenneth French data library (French, n.d.). The New York Stock Exchange (NYSE) calendar spans 2 January 2008 to 2 October 2026, with 550 sessions reserved for estimation before scored returns. Training ends on 31 December 2018, and validation runs from 1 January 2019 to 1 October 2024. The held-out OOS period runs from 2 October 2024 to 2 October 2026, the shorter of the most recent 20% of sessions and two years. The six funds were selected for continuous price and volume histories through the end of the sample; membership is fixed throughout the analysis.

Methodology

At each month-end, estimate daily covariance from 126 completed observations using Ledoit–Wolf shrinkage toward a scaled-identity target (Ledoit and Wolf, 2004). The constrained equal-risk-

contribution (ERC) allocation follows the risk-budgeting framework of Maillard, Roncalli, and Teiletche (2010) and solves

$$\min_b \sum_{i=1}^6 \left[b_i (\widehat{\Sigma}_t b)_i - \frac{b^\top \widehat{\Sigma}_t b}{6} \right]^2, \quad \mathbf{1}^\top b = 1, \quad 0 \leq b_i \leq 0.35.$$

Here b_i is the base weight, $\widehat{\Sigma}_t$ is daily return covariance, and $b_i(\widehat{\Sigma}_t b)_i$ is variance contribution. Binding name caps can prevent exact equality. Shrinkage limits covariance estimation error, which can materially impair optimised allocations out of sample (DeMiguel, Garlappi, and Uppal, 2009).

For current base b_t , calculate historical portfolio returns $x_s = b_t^\top r_s$ using observations through close t . An exponentially weighted moving average (EWMA) with span n applies geometrically declining weights to squared returns. The annualised volatility forecast and capped risky exposure are

$$\widehat{\sigma}_t = \sqrt{252 \max\{\text{EWMA}_{20}(x_s^2), \text{EWMA}_{63}(x_s^2)\}}, \quad k_t = \min(1, 0.10/\widehat{\sigma}_t).$$

Each estimate requires 63 observations. Taking the larger estimate combines a faster response with a slower volatility measure. A five-percentage-point exposure change triggers a trade, while monthly base rebalances proceed independently. Gross exposure cannot exceed one, and the strategy neither shorts nor borrows. The 10% forecast target does not fix realised volatility.

The close- t signal is executed at close $t + 1$, after that session's return; new holdings first earn the next close-to-close return. This convention separates information formation, order execution, and the return earned by the new position. Given pre-fee risky holdings h_i , portfolio net asset value (NAV) V , desired weights w_i , and cost rate c , post-fee NAV satisfies

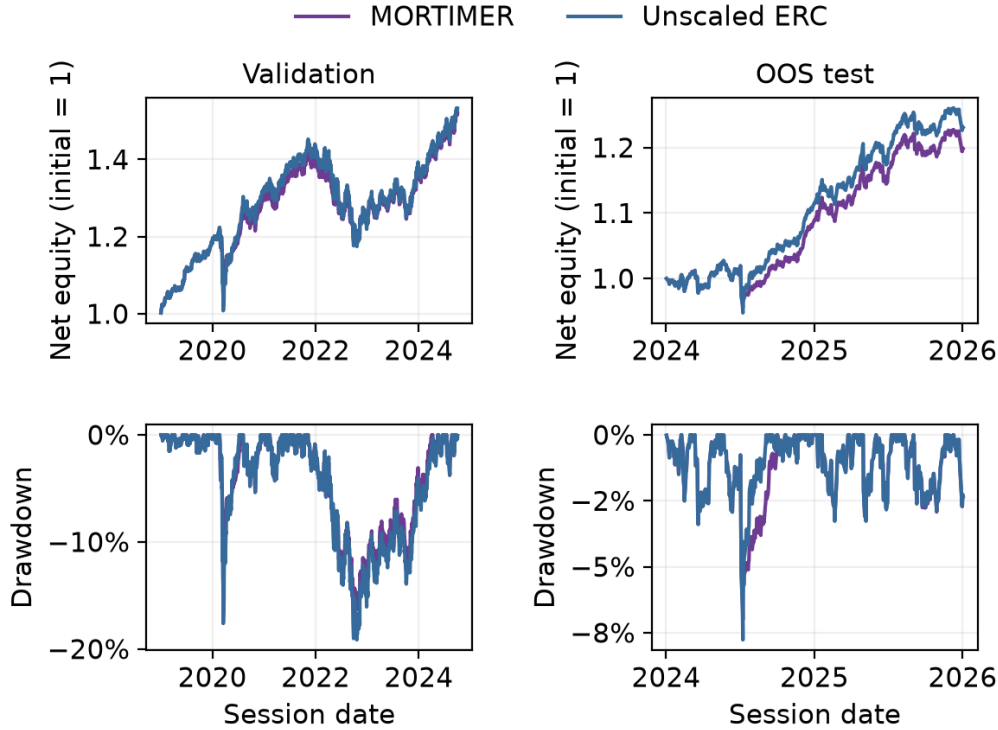
$$V^{post} = V - c \sum_i |V^{post} w_i - h_i|.$$

Costs are five basis points of risky dollars bought plus sold, representing spread, commission, and closing slippage; cash transfers carry no charge. A ten-basis-point cost case is evaluated on development data. The cash return proxy is derived from initial-release three-month Treasury bill yields.

Results

We calculate annualised return as compound annual growth rate (CAGR) from split-local net wealth using 252 trading sessions per year. Annualised volatility is the sample standard deviation of daily returns multiplied by $\sqrt{252}$, and the Sharpe ratio uses daily excess returns relative to the cash return proxy (Sharpe, 1994). Maximum drawdown (MDD) is measured from initial wealth of one. Annual turnover is half the L1 change in risky and cash weights, and transaction costs apply to risky purchases and sales. The table reports all metrics net of five-basis-point costs for each split. Net equity begins at one in each split; the Results figure plots equity and drawdown separately for validation and the held-out test. A separate figure traces rolling volatility, risky exposure, and trailing annual turnover over the full sample, with dashed lines marking the training, validation, and test boundaries. The turnover panel plots a rolling 252-session sum and omits the initial window until a full year of observations is available.

Sample	Rule	Ann. return%	Vol.%	Sharpe	Sortino	MDD%	Calmar	Turnover
Training	MORTIMER	5.85	6.37	0.859	1.232	-12.36	0.473	0.96
Training	Unscaled ERC	6.13	6.51	0.884	1.273	-12.48	0.491	0.45
Validation	MORTIMER	7.56	7.93	0.660	0.916	-16.63	0.455	1.57
Validation	Unscaled ERC	7.71	9.78	0.566	0.781	-19.14	0.403	0.35
OOS test	MORTIMER	9.52	7.47	0.714	0.984	-7.00	1.360	1.12
OOS test	Unscaled ERC	11.00	8.13	0.827	1.181	-7.77	1.416	0.35

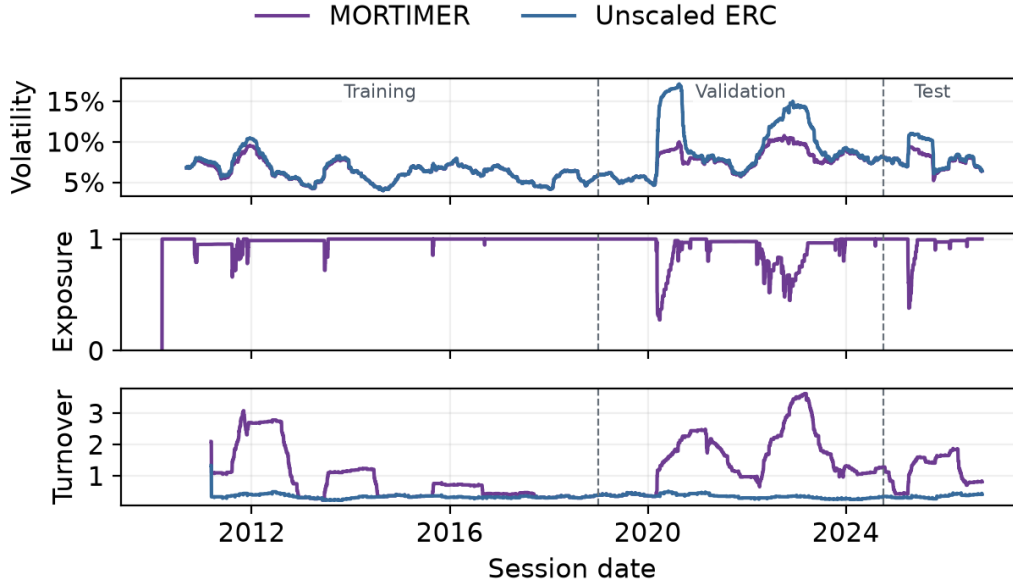


Validation provides the clearest evidence on the overlay’s risk mechanism. Against unscaled ERC, annualised volatility falls from 9.78% to 7.93%, and maximum drawdown narrows from 19.14% to 16.63%. Annualised return is 15 basis points lower. Under the ten-basis-point cost assumption, validation CAGR and Sharpe are 7.46% and 0.649, respectively. A paired 20-session block bootstrap places the validation return difference within a 95% interval of $[-2.94, 2.44]$ percentage points. The interval includes zero, consistent with the modest return difference.

Over the held-out out-of-sample test, annualised volatility and maximum drawdown are lower than unscaled ERC by 8.2% and 9.9%, respectively. The Sharpe ratio is 0.714, compared with 0.827 for the benchmark. This pattern is consistent with the intended reduction in risky exposure, while the validation forecast’s Pearson and rank information coefficients of 0.537 and 0.668 indicate that the volatility signal also orders subsequent realised volatility. Taken together, the results support risk moderation; they do not show improved risk-adjusted performance over unscaled ERC.

Risk management

Base weights are long-only and capped at 35% per instrument. Gross exposure cannot exceed 100%, with the remainder held as cash. Holdings drift between executions, so realised weights can temporarily differ from target weights. The overlay trades when its desired exposure changes by five percentage points; monthly base rebalances remain independent of this threshold. A volatility target controls forecast exposure rather than guaranteeing realised risk, and the lagged signal leaves exposure to shocks until the next execution. Factor attribution on the final common sample estimates market, value, momentum, and Treasury-duration loadings; the coefficients and estimation details are reported in Appendix A.



Liquidity and capacity

Liquidity is measured with previous-session trailing 20-session average daily dollar volume (ADDV). We express portfolio scale as initial assets under management (AUM), equal to net asset value at inception. Participation is $Q/ADDV$, where Q is risky trade notional. The square-root impact scenario follows the empirical market-impact relation documented by Tóth et al. (2011):

$$I = Y\sigma\sqrt{Q/ADDV}, \quad \text{NAV cost} = IQ/NAV,$$

We set $Y = 0.5$ and use lagged volatility σ for each AUM scenario.

Initial AUM (\$m)	Max $Q/ADDV\%$	Impact-net CAGR%
1	0.52	7.55
10	5.22	7.53
100	52.16	7.47
500	260.81	7.35
1000	521.62	7.26

At $Y = 0.5$, maximum participation rises from 0.52% at \$1 million to 5.22% at \$10 million. A 1% participation limit is reached at initial AUM of \$1.92 million. Peak rebalance orders relative to recent volume therefore determine modelled capacity; rows above 100% participation extrapolate the impact formula beyond full daily volume. The square-root specification makes the impact rate increase with the square root of participation, while total impact cost also depends on order notional. Accordingly, the modest change in impact-adjusted return at low portfolio values coexists with a marked increase in peak participation. Annual turnover alone does not describe the liquidity required by individual rebalance orders.

Limitations and further work

The six-fund universe requires continuous price and volume histories, so discontinued funds are absent and historical investability is not fully represented. A point-in-time fund universe would allow membership, closures, and delistings to enter the portfolio as they became observable. Execution data with order-level spreads, commissions, and realised slippage would also permit direct calibration of costs and the impact response across portfolio sizes. These extensions would test whether the measured risk reduction persists under historically investable membership and executable prices.

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A Additional statistical evidence

The volatility forecast is evaluated against realised volatility over the following 21 sessions. For forecast $\hat{v}_t$ and subsequent realised measure $v_{t+1:t+21}$, Pearson information coefficient (IC) is their sample correlation; rank IC is the correlation of their cross-time ranks. On the validation sample, Pearson IC is 0.537 and rank IC is 0.668 over 1,426 forecast dates. These values quantify the ordering of subsequent volatility by the forecast and are separate from the return comparison in the Results section.

Return-difference uncertainty is estimated with a paired moving-block bootstrap. The procedure resamples matched daily returns in 20-session blocks, preserving local dependence and the pairing between MORTIMER and unscaled ERC. Across 1,000 resamples, the validation annualised-return difference has a 95% interval of $[-2.94, 2.44]$ percentage points. The interval spans zero, so the validation sample does not resolve the sign of the mean return difference at this confidence level.

B Factor attribution

The regression uses daily net portfolio returns in excess of the Kenneth French risk-free series and five factors: market, size, value, momentum, and Treasury-duration returns. Newey–West standard errors use five lags. Over 479 common held-out observations through 31 August 2026, estimated coefficients are 0.272 for market, 0.071 for size, 0.031 for value, 0.066 for momentum, and 0.198 for duration. The daily intercept is 0.000098 with a heteroskedasticity-and-autocorrelation-consistent t -statistic of 0.69. These loadings describe broad return exposures; the equity factors do not span the full set of cross-asset value and momentum risks.

C Capacity model

The capacity calculation scales the initial portfolio value while holding the observed validation trading schedule fixed. For each risky order, participation is $Q/ADDV$, with average daily dollar volume measured over the preceding 20 sessions. The square-root impact rate $I = Y\sigma\sqrt{Q/ADDV}$ is applied to risky notional, using $Y = 0.5$ and lagged volatility. Under these assumptions, maximum single-order participation is 0.52% at \$1 million initial NAV, 5.22% at \$10 million, and 52.16% at \$100 million; the corresponding impact-adjusted validation CAGR is 7.55%, 7.53%, and 7.47%. At larger values the model exceeds full daily volume, so those rows are mathematical extrapolations rather than plausible execution schedules.